Automatic Number Plate Recognition (ANPR)

Internship Project Report

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Introduction

Automatic Number Plate Recognition (ANPR) is an advanced application of computer vision and machine learning that has gained significant importance in recent years due to the rapid growth of vehicular traffic and the increasing need for intelligent monitoring systems. ANPR systems are designed to automatically capture the license plate of a vehicle and extract the alphanumeric characters for identification purposes. This technology is widely applied in traffic surveillance, law enforcement, toll collection, parking management, and security monitoring.

The significance of ANPR lies in its ability to automate vehicle tracking and monitoring without the need for manual intervention. Traditional methods of recording and checking vehicles require human effort, which is both time-consuming and error-prone. In contrast, ANPR provides a real-time, scalable, and efficient solution for identifying vehicles in urban environments.

Technically, ANPR is implemented through a two-step process. First, an object detection algorithm locates and crops the license plate from a vehicle image. Second, an Optical Character Recognition (OCR) module processes the cropped image and extracts the alphanumeric sequence. In this project, YOLOv8 (You Only Look Once), a state-of-the-art object detection algorithm, is used for license plate detection. For OCR, libraries such as EasyOCR and PaddleOCR were employed, as they provide robust recognition under challenging conditions such as varying lighting, blur, and plate styles.

The project was developed using Python, leveraging OpenCV for image preprocessing, YOLOv8 for detection, and OCR libraries for text recognition. The system was tested on Indian number plates, which present additional challenges due to variability in design, font, and layout compared to standardized plates in other countries.

This internship project allowed the exploration of deep learning, computer vision, and intelligent transport applications. Beyond technical knowledge, it demonstrated how artificial intelligence can contribute to real-world urban infrastructure problems. With proper optimization and scaling, ANPR systems can significantly improve traffic management, law enforcement, and automated access control, making cities smarter and safer.

Project Summary

The Automatic Number Plate Recognition (ANPR) project undertaken during my internship was designed to develop a complete pipeline that combines number plate detection with text recognition for accurate vehicle identification. The project focused on Indian number plates, which are more diverse and complex than those in many countries due to variations in font, size, background color, and state-specific formats.

The pipeline began with number plate detection using YOLOv8, a cutting-edge deep learning model known for its real-time object detection capabilities. YOLOv8 was trained on a dataset of annotated Indian license plate images. During inference, YOLOv8 scanned each frame from a video stream, identified the license plate region, and generated bounding boxes around it. These cropped regions were then passed to the next stage of processing.

For character recognition, Optical Character Recognition (OCR) libraries such as EasyOCR and PaddleOCR were tested. These libraries applied deep learning-based text recognition to extract the alphanumeric characters from the cropped plate region. Since OCR often faces difficulties due to poor lighting, reflections, or motion blur, preprocessing steps such as grayscale conversion, noise reduction, and contrast enhancement were applied to improve recognition quality.

The system was built in Python, with OpenCV handling image and video preprocessing, Ultralytics YOLOv8 managing detection, and OCR libraries performing text extraction. The entire workflow was executed on Google Colab, which provided GPU acceleration to handle the training and inference tasks efficiently.

To validate the performance of the system, multiple experiments were conducted with both images and videos of vehicles. The results were encouraging, with the system capable of detecting plates with high accuracy and recognizing most alphanumeric sequences correctly. The recognized results were displayed on the video feed in real time and simultaneously stored in a structured database for record-keeping and analysis.

In summary, this project successfully delivered a functional ANPR system with real-time detection and recognition capabilities

System Requirements

For the successful design and implementation of the Automatic Number Plate Recognition (ANPR) system, certain hardware and software requirements had to be met. These requirements were carefully considered to ensure that the system could operate efficiently during both the training phase and real-time deployment.

Hardware Requirements:

1. Processor (CPU): A minimum quad-core processor was required to handle video processing and model inference. Higher-end CPUs improved processing speed.
2. Graphics Processing Unit (GPU): Since YOLOv8 is a deep learning model, GPU acceleration was necessary for training and real-time detection. NVIDIA GPUs with CUDA support (such as RTX 3060/3080) significantly improved performance.
3. Memory (RAM): At least 8 GB of RAM was required, though 16 GB or more was recommended to manage large datasets, real-time video streams, and simultaneous OCR operations.
4. Storage: A minimum of 50 GB was necessary to store datasets, trained weights, video files, and output results.
5. Camera/Video Input Device: For real-time testing, a camera with at least 720p resolution was recommended.

Software Requirements:

1. Programming Language: Python 3.8 or higher was used due to its vast ecosystem of libraries for machine learning and computer vision.
2. Frameworks & Libraries:
   * OpenCV: For handling image preprocessing and video operations.
   * Ultralytics YOLOv8: For number plate detection.
   * EasyOCR / PaddleOCR: For character recognition.
   * NumPy and Pandas: For data handling.
   * Matplotlib/Seaborn: For visualization and result analysis.
3. Development Environment: Google Colab and Jupyter Notebook were used for training and testing, while VS Code was used for debugging and script execution.

4. Operating System: Windows or Linux with Python and GPU driver support.

Dataset Requirements: The dataset needed annotated images of Indian number plates in YOLO format. Each image required a corresponding .txt file containing bounding box coordinates. Data augmentation was applied to increase dataset diversity and robustness.

Methodology / Algorithm

Step 1: Data Collection & Annotation

The first step involved collecting a dataset of Indian vehicles with visible license plates. Images were sourced from open datasets and custom vehicle footage. Annotation was carried out in YOLO format, where each license plate was manually marked with a bounding box. This dataset was then split into training, validation, and testing subsets.

Step 2: Preprocessing

Data preprocessing was performed to improve detection and recognition performance. Techniques such as image resizing, normalization, grayscale conversion, and histogram equalization were applied.

Step 3: Number Plate Detection (YOLOv8)

YOLOv8, a state-of-the-art object detection algorithm, was trained on the annotated dataset. During inference, YOLOv8 scanned each video frame, identified the number plate, and generated a bounding box around it. This cropped plate image was passed on for character recognition. YOLOv8 was chosen for its speed and accuracy, making it suitable for real-time applications.

Step 4: Optical Character Recognition (OCR)

Once the number plate region was detected, it was cropped and processed for OCR. EasyOCR and PaddleOCR libraries were used to extract alphanumeric characters. To enhance recognition accuracy, preprocessing steps such as thresholding, denoising, and contrast adjustment were applied. The extracted text was cleaned and formatted into standard license plate patterns.

Step 5: Post-Processing & Output

The recognized license plate number was displayed in real time on the video feed. Additionally, results were stored in structured files (CSV/database) for further analysis and record-keeping.

This methodology ensured that the ANPR system achieved a balance between detection accuracy and processing speed, enabling it to function effectively in real-time environments.

SnapShots

Fig 1 - Number Plate recognition

Fig 2- Number Plate recognition

Fig 3 -Vehicle detection

Challenges Faced

During the development of the Automatic Number Plate Recognition (ANPR) project, several challenges were encountered, both technical and practical. These challenges provided valuable learning experiences and highlighted areas for improvement in real-world deployments.

1. Variation in Number Plates: Unlike many countries with standardized plate formats, Indian number plates vary widely in font style, size, color, and alignment. This inconsistency reduced the accuracy of both YOLO detection and OCR recognition. Plates with decorative fonts or regional language characters posed additional difficulties.

2. Lighting Conditions: Poor lighting, strong reflections, and glare from headlights or streetlights affected the clarity of captured images. Plates photographed at night or under shadows were harder for OCR to interpret accurately.

3. Motion Blur: Vehicles moving at high speed often produced blurred license plate regions, making both detection and text extraction less reliable. Although preprocessing techniques were applied, results were not always satisfactory.

4. Dataset Limitations: Publicly available datasets for Indian license plates were limited in size and diversity. Additional manual annotation was required, which was time-consuming and labor-intensive.

5. OCR Inaccuracies: OCR frequently confused visually similar characters (e.g., 0 with O, 1 with I, and 5 with S). This reduced recognition accuracy, especially when plate images were of low resolution.

6. Real-Time Performance: Achieving a balance between accuracy and speed was challenging. While YOLOv8 provided high detection accuracy, running both detection and OCR in real-time required significant computational power.

7. Environmental Noise: Dust, dirt, or obstructions on plates made recognition harder. Vehicles with damaged or intentionally tampered plates also posed recognition challenges.

Despite these obstacles, iterative improvements were made. Data augmentation increased robustness, advanced preprocessing improved OCR results, and GPU acceleration enhanced performance. These challenges reflected real-world complexities and underscored the importance of continuous refinement in ANPR systems.

Future Enhancements

While the developed ANPR system achieved promising results, several enhancements can be made to improve accuracy, scalability, and real-world usability.

1. Advanced OCR Models: Current OCR systems sometimes misinterpret characters. Integrating transformer-based OCR models such as TrOCR or Donut can significantly improve recognition performance under noisy conditions.

2. Multi-Language Support: Many Indian number plates include text in regional languages in addition to English. Enhancing OCR to support multi-language recognition would make the system more versatile.

3. Edge Deployment: Currently, the system requires GPU support for real-time operation. Optimizing the pipeline for lightweight edge devices like Raspberry Pi or Jetson Nano would allow deployment in roadside units or CCTV systems.

4. Database Integration: Linking the ANPR system to national vehicle registration databases would enable automatic verification of vehicle details, detection of stolen vehicles, and identification of traffic violators.

5. Fraud Detection: Some vehicles use fake or tampered plates to evade law enforcement. Incorporating fraud detection mechanisms using image forensics or anomaly detection could address this issue.

6. Cloud and IoT Integration: Deploying ANPR in cloud-based environments would enable city-wide traffic monitoring. IoT-enabled cameras could feed data to a centralized system for large-scale analysis.

7. Multi-Camera Tracking: Future versions could support vehicle tracking across multiple surveillance cameras, providing authorities with movement patterns and location histories.

8. Scalability: Improving the system to handle high volumes of vehicles simultaneously in congested traffic conditions would make it more practical for smart city applications.

These enhancements would transform the system from a standalone prototype into a fully deployable intelligent transport solution, capable of addressing challenges in law enforcement, traffic regulation, and smart urban planning.

Conclusion

The Automatic Number Plate Recognition (ANPR) internship project successfully demonstrated the integration of computer vision and deep learning techniques to create a real-time vehicle identification system. By using YOLOv8 for license plate detection and OCR libraries for character recognition, the system was able to process images and video streams with promising accuracy.

Although the project faced challenges such as plate variability, lighting conditions, and OCR misclassifications, iterative improvements ensured that the pipeline performed reliably in most scenarios. The ability to detect and recognize license plates in real time highlights the potential of ANPR systems in traffic management, parking automation, toll collection, and law enforcement.

From a learning perspective, this internship provided valuable exposure to modern deep learning workflows, dataset annotation, and system optimization. Skills gained included working with YOLO models, applying image preprocessing techniques, implementing OCR, and handling real-time video processing. Beyond technical growth, the project emphasized problem-solving, adaptability, and the importance of designing AI solutions for real-world challenges.

Looking forward, ANPR has the potential to become an essential component of smart city infrastructure. With enhancements such as multi-language OCR, cloud deployment, and fraud detection, the system could scale to city-wide deployments, aiding governments and private organizations alike.

In conclusion, this project not only served as a successful internship outcome but also as a foundation for future innovations in intelligent transportation systems. It demonstrates how artificial intelligence, when combined with practical engineering, can create impactful solutions for modern society.